



LABORATORY REPORT



Name : Mrs SWETAL SHAH	Sex/Age : Female / 48 Years	Case ID : 40300121703
Ref. By : DR. HETAN SHAH	Dis. At :	Pt. ID : 3440028
Bill. Loc. :		Pt. Loc. :
Reg Date and Time : 18-Mar-2024 09:59	Sample Type :	Mobile No. : 7819811953
Sample Date and Time : 18-Mar-2024 09:13	Sample Coll. By : NSRL	Ref Id1 :
Report Date and Time :	Acc. Remarks : -	Ref Id2 :

Abnormal Result(s) Summary

Test Name	Result Value	Unit	Reference Range
Glyco Hemoglobin (HbA1c)			
HbA1C	5.90	% of total Hb	<5.7: Normal 5.7-6.4: Prediabetes >=6.5: Diabetes
Haemogram (CBC)			
Neutrophil	70.2	%	40.00 - 70.00
Lymphocyte	19.0	%	20.00 - 40.00
Eosinophil	517	/μL	20.00 - 500.00
Neut/Lympho Ratio (NLR)	3.69		0.78 - 3.53
Lipid Profile			
HDL Cholesterol	28.00	mg/dL	40 - 60
Triglyceride	169.00	mg/dL	<150
Chol/HDL	6.75		0 - 4.1
LDL Cholesterol	127.20	mg/dL	0.00 - 100.00
Liver Function Test			
S.G.P.T.	39.00	U/L	< 34
Renal Function Test			
Urea	11.00	mg/dL	14.98 - 40.01
Urine Examination			
Leucocytes (ESTERASE)	+++		Negative
Blood	+		Negative
Leucocyte	171	/HPF	Nil
RBC (Isomorphic)	21	/HPF	Nil

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

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25 OH Cholecalciferol (D2+D3) **20.60** ng/mL

Adult:
 30 - 100 Normal Level
 20 - <30 Insufficiency
 < 20 Deficiency
 > 150 Toxicity

Vitamin B - 12 Level **1481.00** pg/mL

Pediatric
 20 - 100 Normal Level
 15 - <20 Insufficiency
 < 15 Deficiency
 187 - 883

Abnormal Result(s) Summary End

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TEST	RESULTS	UNIT	BIOLOGICAL REF. INTERVAL	REMARKS
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HAEMOGRAM REPORT

HB AND INDICES

Haemoglobin	13.2	G%	12.00 - 15.00
RBC (Electrical Impedance)	4.41	millions/cumm	3.80 - 4.80
PCV(Calc)	41.10	%	36.00 - 46.00
MCV (RBC histogram)	93.2	fL	83.00 - 101.00
MCH (Calc)	29.9	pg	27.00 - 32.00
MCHC (Calc)	32.1	gm/dL	31.50 - 34.50
RDW (RBC histogram)	13.80	%	11.00 - 16.00

TOTAL AND DIFFERENTIAL WBC COUNT (Flowcytometry)

Total WBC Count	9230	/μL	4000.00 - 10000.00
Neutrophil	H [%] 70.2	%	EXPECTED VALUES 40.00 - 70.00 [Abs] 6479 /μL EXPECTED VALUES 2000.00 - 7000.00
Lymphocyte	L 19.0	%	20.00 - 40.00 1754 /μL 1000.00 - 3000.00
Eosinophil	5.6	%	1.00 - 6.00 H 517 /μL 20.00 - 500.00
Monocytes	4.8	%	2.00 - 10.00 443 /μL 200.00 - 1000.00
Basophil	0.4	%	0.00 - 2.00 37 /μL 0.00 - 100.00
Neut/Lympho Ratio (NLR)	H 3.69		0.78 - 3.53

PLATELET COUNT (Optical)

Platelet Count	356000	/μL	150000.00 - 410000.00
MPV	10.10	fL	6.5 - 12
PDW	15.6	fL	15 - 17

SMEAR STUDY

RBC Morphology	Normocytic Normochromic RBCs.
WBC Morphology	Eosinophilia

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr. Aakash Shah

M.D. (Pathologist)

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Dr. Sandip Shah

M.D. (Path. & Bact.)

Consultant Pathologist

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Laboratory : "KEDAR" Opposite Krupa Petrol Pump, Near Parimal Garden, Ahmedabad - 380006 ☎ 079-40408181 / 61618181

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Regd. Office : Plot No. 7, Industrial Estate, Rajiv Gandhi Salai, Perungudi, Chennai - 600096, Tamil Nadu, India. | CIN - U85300TN2017PTC114099

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Platelet

Platelets are adequate in number.

Parasite

Malarial Parasite not seen on smear.

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Sample Date and Time : 18-Mar-2024 09:13	Sample Coll. By : NSRL	Ref Id1 :
Report Date and Time : 18-Mar-2024 10:35	Acc. Remarks : -	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
ESR Photometrical capillary stopped flow kinetic analysis	19	mm after 1hr	3 - 20	

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Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist
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Sample Date and Time : 18-Mar-2024 09:13	Sample Coll. By : NSRL	Ref Id1 :
Report Date and Time : 18-Mar-2024 10:57	Acc. Remarks : -	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
Plasma Glucose - F <i>Photometric, Hexokinase</i>	76.00	mg/dL	70.0 - 100	

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Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist
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MD. Path.
Consultant Pathologist

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TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
<u>Glycated Haemoglobin Estimation</u>				
HbA1C HPLC	H 5.90	% of total Hb	<5.7: Normal 5.7-6.4: Prediabetes >=6.5: Diabetes	
Estimated Avg Glucose (3 Mths) Calculated	122.63	mg/dL	Not available	

Please Note change in reference range as per ADA 2021 guidelines.

Interpretation :

HbA1C level reflects the mean glucose concentration over previous 8-12 weeks and provides better indication of long term glycemic control.

Levels of HbA1C may be low as result of shortened RBC life span in case of hemolytic anemia.

Increased HbA1C values may be found in patients with polycythemia or post splenectomy patients.

Patients with Homozygous forms of rare variant Hb(CC,SS,EE,SC) HbA1c can not be quantitated as there is no HbA.

In such circumstances glycemic control can be monitored using plasma glucose levels or serum Fructosamine.

The A1c target should be individualized based on numerous factors, such as age, life expectancy, comorbid conditions, duration of diabetes, risk of hypoglycemia or adverse consequences from hypoglycemia, patient motivation and adherence.

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Consultant Pathologist
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MD. Path.
Consultant Pathologist

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TEST	RESULTS	UNIT BIOLOGICAL REF RANGE	REMARKS
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BIOCHEMICAL INVESTIGATIONS

Renal Function Test

Urea <i>Urease</i>	L	11.00	mg/dL	14.98 - 40.01
Creatinine <i>Jaffe - Kinetic</i>		0.60	mg/dL	0.5 - 1.1
Uric Acid <i>Uricase</i>		5.50	mg/dL	2.6 - 6.0
Sodium <i>ISE</i>		141.00	mmol/L	136 - 145
Potassium <i>ISE</i>		4.30	mmol/L	3.5 - 5.1
Chloride <i>ISE</i>		107.00	mmol/L	98 - 107

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Consultant Pathologist
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TEST	RESULTS	UNIT BIOLOGICAL REF RANGE	REMARKS
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BIOCHEMICAL INVESTIGATIONS

Liver Function Test

S.G.P.T. <i>NADH (Without P-5-P)</i>	H	39.00	U/L	< 34
S.G.O.T. <i>NADH (Without P-5-P)</i>		31.00	U/L	5.0 - 34.0
Alkaline Phosphatase <i>Enzymatic, PNPP-AMP</i>		66.00	U/L	46 - 122
Gamma Glutamyl Transferase <i>Multipoint Rate-L-y-Glytamyl-p-nitroanilide</i>		36.00	U/L	0 - 38
Proteins (Total) <i>Biuret</i>		7.40	gm/dL	6.40 - 8.30
Albumin <i>(BCG)</i>		4.30	gm/dL	3.5 - 5.2
Globulin		3.10	gm/dL	2 - 4.1
A/G Ratio <i>CALC</i>		1.4		1 - 2.1
Bilirubin Total <i>Colorimetric Diazo Method</i>		0.60	mg/dL	0.3 - 1.2
Bilirubin Conjugated <i>Colorimetric Diazo Method</i>		0.20	mg/dL	0 - 0.50
Bilirubin Unconjugated		0.40	mg/dL	0 - 0.8

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M.D. (Path. & Bact.)
Consultant Pathologist
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BIOCHEMICAL INVESTIGATIONS

Skeletal Profile

Calcium <i>OCPC</i>	9.00	mg/dL	8.4 - 10.2	
Phosphorus Inorganic <i>Phosphomolybdate</i>	2.90	mg/dL	2.3 - 4.7	

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TEST	RESULTS	UNIT BIOLOGICAL REF RANGE	REMARKS
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BIOCHEMICAL INVESTIGATIONS

Lipid Profile

Cholesterol <i>Enzymatic</i>	189.00	mg/dL	110 - 200
HDL Cholesterol <i>Accelerator Selective Detergent</i>	L 28.00	mg/dL	40 - 60
Triglyceride <i>Arsenazo - Colorimetric</i>	H 169.00	mg/dL	<150
VLDL	33.80	mg/dL	10 - 40
Chol/HDL <i>Calculated</i>	H 6.75		0 - 4.1
LDL Cholesterol <i>Enzymatic</i>	H 127.20	mg/dL	0.00 - 100.00

NEW ATP III GUIDELINES (MAY 2001). MODIFICATION OF NCEP

LDL CHOLESTEROL	CHOLESTEROL	HDL CHOLESTEROL	TRIGLYCERIDES
Optimal<100	Desirable<200	Low<40	Normal<150
Near Optimal 100-129	Border Line 200-239	High >60	Border High 150-199
Borderline 130-159	High >240	-	High 200-499
High 160-189	-	-	-

- LDL Cholesterol level is primary goal for treatment and varies with risk category and assesment
- For LDL Cholesterol level Please consider direct LDL value
Risk assesment from HDL and Triglyceride has been revised. Also LDL goals have changed.
- Detail test interpretation available from the lab
- All tests are done according to NCEP guidelines and with FDA approved kits.
- LDL Cholesterol level is primary goal for treatment and varies with risk category and assesment

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M.D. (Pathologist)

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Sample Date and Time : 18-Mar-2024 09:13 Sample Coll. By : NSRL Ref Id1 :
Report Date and Time : 18-Mar-2024 12:26 Acc. Remarks : - Ref Id2 :

TEST	RESULTS	UNIT	BRI	TEST	RESULTS	UNIT	BRI
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URINE EXAMINATION (STRIP METHOD AND AUTOMATED IMAGE EVALUATION)

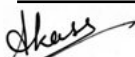
Physical and Chemical examination

Colour	Light Yellow			Glucose	Negative		Negative
Transparency	Very turbid			Ketone Bodies Urine	Negative		Negative
Sp.Gravity	1.014	1.003 - 1.035		Urobilinogen	Negative		Negative
pH	5.50	4.6 - 8		Bilirubin	Negative		Negative
Leucocytes (ESTERASE)	+++	Negative		Nitrite	Negative		Negative
Protein	Negative	Negative		Blood	+		Negative

Automated Microscopy

Leucocyte	171	/HPF	Nil	Granular Cast	Nil	/HPF	Nil
Leucocyte Clumps	2	/HPF		RBC cast	Nil	/HPF	Nil
RBC (Isomorphic)	21	/HPF	Nil	WBC Cast	Nil	/HPF	Nil
RBC (Dysmorphic)	Nil	/HPF	Nil	Fatty Cast	Nil	/HPF	Nil
<u>Epithelial cells</u>				Mixed Cast	Nil	/HPF	Nil
Squamous EC	144	/HPF	Present	<u>Crystals</u>			
NonSquamous EC	Nil	/HPF		Crystals	5	/HPF	Nil
Transitional EC	Nil	/HPF		Calcium oxalate Monohydrate	Nil	/HPF	Nil
Renal Tubular EC	Nil	/HPF		Calcium oxalate Dihydrate	2	/HPF	Nil
<u>Bacteria</u>				Triple phosphate	Nil	/HPF	Nil
Bacteria	2218	/μL	Nil	Uric acid	Nil	/HPF	Nil
Bacteria Rods	Nil	/μL	Nil	Leucine	Nil	/HPF	Nil
Bacteria Cocci	1888	/μL	Nil	Amorphous	Nil	/HPF	Nil
Yeast	Nil	/μL	Nil	Trichomonas	Nil	/μL	Nil
<u>Cast</u>							
Hyaline Cast	Nil	/HPF	Nil				
Pathological Cast	Nil	/HPF	Nil				
Hyaline granular cast	Nil	/HPF	Nil				

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TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
Thyroid Function Test				
Triiodothyronine (T3) CMIA	1.05	ng/mL	0.64 - 1.52	
Thyroxine (T4) CMIA	9.22	µg/dL	4.87 - 11.72	
TSH CMIA	2.40	µIU/mL	0.35 - 4.94	

INTERPRETATIONS

- Circulating TSH measurement has been used for screening for euthyroidism, screening and diagnosis for hyperthyroidism & hypothyroidism. Suppressed TSH ($<0.01 \mu\text{IU/mL}$) suggests a diagnosis of hyperthyroidism and elevated concentration ($>7 \mu\text{IU/mL}$) suggest hypothyroidism. TSH levels may be affected by acute illness and several medications including dopamine and glucocorticoids. Decreased (low or undetectable) in Graves disease. Increased in TSH secreting pituitary adenoma (secondary hyperthyroidism), PRTH and in hypothalamic disease thyrotropin (tertiary hyperthyroidism). Elevated in hypothyroidism (along with decreased T4) except for pituitary & hypothalamic disease.
- Mild to modest elevations in patient with normal T3 & T4 levels indicates impaired thyroid hormone reserves & incipient hypothyroidism (subclinical hypothyroidism).
- Mild to modest decrease with normal T3 & T4 indicates subclinical hyperthyroidism.
- Degree of TSH suppression does not reflect the severity of hyperthyroidism, therefore, measurement of free thyroid hormone levels is required in patient with a suppressed TSH level.

CAUTIONS

Sick, hospitalized patients may have falsely low or transiently elevated thyroid stimulating hormone. Some patients who have been exposed to animal antigens, either in the environment or as part of treatment or imaging procedure, may have circulating antianimal antibodies present. These antibodies may interfere with the assay reagents to produce unreliable results.

TSH ref range in pregnancy

First trimester
Second trimester
Third trimester

Reference range (microIU/ml)

0.24 - 2.00
0.43-2.2
0.8-2.5

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist
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Dr. Aakash Shah
MD. Path.
Consultant Pathologist

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

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Neuberg Diagnostics Private Limited

Laboratory : "KEDAR" Opposite Krupa Petrol Pump, Near Parimal Garden,
Ahmedabad - 380006 ☎ 079-40408181 / 61618181
✉ contact@neubergsupratech.com

Regd. Office : Plot No. 7, Industrial Estate, Rajiv Gandhi Salai, Perungudi,
Chennai - 600096, Tamil Nadu, India. | CIN - U85300TN2017PTC114099
🌐 www.neubergsupratech.com



MC-6136

LABORATORY REPORT



Name : Mrs SWETAL SHAH	Sex/Age : Female / 48 Years	Case ID : 40300121703
Ref. By : DR. HETAN SHAH	Dis. At :	Pt. ID : 3440028
Bill. Loc. :		Pt. Loc :
Reg Date and Time : 18-Mar-2024 09:59	Sample Type : Serum	Mobile No. : 7819811953
Sample Date and Time : 18-Mar-2024 09:13	Sample Coll. By : NSRL	Ref Id1 :
Report Date and Time : 18-Mar-2024 11:22	Acc. Remarks : -	Ref Id2 :

Interpretation Note:

Ultra sensitive-thyroid-stimulating hormone (TSH) is a highly effective screening assay for thyroid disorders. In patients with an intact pituitary-thyroid axis, s-TSH provides a physiologic indicator of the functional level of thyroid hormone activity. Increased s-TSH indicates inadequate thyroid hormone, and suppressed s-TSH indicates excess thyroid hormone. Transient s-TSH abnormalities may be found in seriously ill, hospitalized patients, so this is not the ideal setting to assess thyroid function. However, even in these patients, s-TSH works better than total thyroxine (an alternative screening test). When the s-TSH result is abnormal, appropriate follow-up tests T4 & free T3 levels should be performed. If TSH is between 5.0 to 10.0 & free T4 & free T3 level are normal then it is considered as subclinical hypothyroidism which should be followed up after 4 weeks & If TSH is > 10 & free T4 & free T3 level are normal then it is considered as overt hypothyroidism.

Serum triiodothyronine (T3) levels often are depressed in sick and hospitalized patients, caused in part by the biochemical shift to the production of reverse T3. Therefore, T3 generally is not a reliable predictor of hypothyroidism. However, in a small subset of hyperthyroid patients, hyperthyroidism may be caused by overproduction of T3 (T3 toxicosis). To help diagnose and monitor this subgroup, T3 is measured on all specimens with suppressed s-TSH and normal FT4 concentrations.

Normal ranges of TSH & thyroid hormones vary according to trimester in pregnancy.

TSH ref range in Pregnancy	Reference range (microIU/ml)
First trimester	0.24 - 2.00
Second trimester	0.43-2.2
Third trimester	0.8-2.5

	T3	T4	TSH
Normal Thyroid function	N	N	N
Primary Hyperthyroidism	↑	↑	↓
Secondary Hyperthyroidism	↑	↑	↑
Grave's Thyroiditis	↑	↑	↑
T3 Thyrotoxicosis	↑	N	N/↓
Primary Hypothyroidism	↓	↓	↑
Secondary Hypothyroidism	↓	↓	↓
Subclinical Hypothyroidism	N	N	↑
Patient on treatment	N	N/↑	↓

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)



Dr. Sandip Shah

M.D. (Path. & Bact.)
Consultant Pathologist
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MD. Path.
Consultant Pathologist

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TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
25 OH Cholecalciferol (D2+D3) CMIA	L 20.60	ng/mL	Adult: 30 - 100 Normal Level 20 - <30 Insufficiency < 20 Deficiency > 150 Toxicity Pediatric 20 - 100 Normal Level 15 - <20 Insufficiency < 15 Deficiency	

25-OH-VitD plays a primary role in the maintenance of calcium homeostasis. It promotes intestinal calcium absorption and, in concert with PTH, skeletal calcium deposition, or less commonly, calcium mobilization. Modest 25-OH-VitD deficiency is common; in institutionalised elderly, its prevalence may be >50%. Although much less common, severe deficiency is not rare either. Reasons for suboptimal 25-OH-VitD levels include lack of sunshine exposure, a particular problem in Northern latitudes during winter; inadequate intake; malabsorption (e.g. due to Celiac disease); depressed hepatic vitamin D 25-hydroxylase activity, secondary to advanced liver disease; and enzyme-inducing drugs, in particular many antiepileptic drugs, including phenytoin, phenobarbital, and carbamazepine, that increase 25-OH-VitD metabolism. Hypervitaminosis D is rare, and is only seen after prolonged exposure to extremely high doses of vitamin D. When it occurs, it can result in severe hypercalcemia and hyperphosphatemia.

INTERPRETATION

- Levels <10 ng/mL may be associated with more severe abnormalities and can lead to inadequate mineralization of newly formed osteoid, resulting in rickets in children and osteomalacia in adults. In these individuals, serum calcium levels may be marginally low, and parathyroid hormone (PTH) and serum alkaline phosphatase are usually elevated. Definitive diagnosis rests on the typical radiographic findings or bone biopsy/histomorphometry.
- Patients who present with hypercalcemia, hyperphosphatemia, and low PTH may suffer either from ectopic, unregulated conversion of 25-OH-VitD to 1,25 (OH)₂-VitD, as can occur in granulomatous diseases, particularly sarcoidosis, or from nutritionally-induced hypervitaminosis D. Serum 1,25 (OH)₂-VitD levels will be high in both groups, but only patients with hypervitaminosis D will have serum 25-OH-VitD concentrations of >80 ng/mL, typically >150 ng/mL.
- Patients with CKD have an exceptionally high rate of severe vitamin D deficiency that is further exacerbated by the reduced ability to convert 25-OH-VitD into the active form, 1,25 (OH)₂-VitD. Emerging evidence also suggests that the progression of CKD & many of the cardiovascular complications may be linked to hypovitaminosis D.
- Approximately half of Stage 2 and 3 CKD patients are nutritional vitamin D deficient (25-OH-VitD, less than 30 ng/mL), and this deficiency is more common among stage 4 CKD patients. Additionally, calcitriol (1,25 (OH)₂-VitD) levels are also overtly low (less than 22 pg/mL) in CKD patients. Similarly, vast majority of dialysis patients are found to be deficient in nutritional vitamin D and have low calcitriol levels. Recent data suggest an elevated PTH is a poor indicator of deficiencies of nutritional vitamin D and calcitriol in CKD patients. CAUTIONS Long term use of anticonvulsant medications may result in vitamin D deficiency that could lead to bone disease; the anticonvulsants most implicated are phenytoin, phenobarbital, carbamazepine, and valproic acid.

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

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TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
VITAMIN B - 12				
Vitamin B - 12 Level CMIA	H 1481.00	pg/mL	187 - 883	

Introduction :

Vitamin B12, a member of the corrin family, is a cofactor for the formation of myelin, and along with folate, is required for DNA synthesis. Levels above 300 or 400 are rarely associated with B12 deficiency induced hematological or neurological disease.

Clinical Significance :

Causes of Vitamin B12 deficiency can be divided into three classes: Nutritional, malabsorption syndromes and gastrointestinal causes. B12 deficiency can cause Megaloblastic anemia (MA), nerve damage and degeneration of the spinal cord. Lack of B12 even mild deficiencies damages the myelin sheath. The nerve damage caused by a lack of B12 may become permanently debilitating.

The relationship between B12 and MA is not always clear that some patients with MA will have normal B12 levels; conversely, many individuals with B12 deficiency are not afflicted with MA.

Decreased in:

Iron deficiency, normal near-term pregnancy, vegetarianism, partial gastrectomy/ileal damage, celiac disease, use of oral contraception, parasitic competition, pancreatic deficiency, treated epilepsy and advancing age.

Increased in:

Renal failure, liver disease and myeloproliferative diseases.

Variations due to age Increases: with age.

Temporarily Increased after Drug.

Falsely high in Deteriorated sample.

----- End Of Report -----

For test performed on specimens received or collected from non-NSRL locations, it is presumed that the specimen belongs to the patient named or identified as labeled on the container/test request and such verification has been carried out at the point generation of the said specimen by the sender. NSRL will be responsible Only for the analytical part of test carried out. All other responsibility will be of referring Laboratory.

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

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